# DICTO ATOM-13×13

Thirteen clusters of thirteen: twelve around one, six golden and seven periodic, with the pieces that fill between them

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Version 1 · 13 October 2026 · DOI 10.5281/zenodo.23333184 · Licence CC BY 4.0 · Data, scripts, renders and checks: papers/thirteens, docs/PLAN-13.md and studies/2026-10-13-rd13-block at https://github.com/DICTOR-Master/kaleidohedra.
The clusters and their names (Lotus seed, DESHI, UNITY, VENUS, DICTO-Star, Bodhi seed, DICTO Spheroid-13 RD, the DICTO Jewel, the truncated octahedron's 4 + 5 + 4 with its two square windows, the piece names, DICTO ATOM-13×13), the choice of members and the gold centre are DICTO's. Computation and checking with Claude (Anthropic).

**Abstract.** Twelve equal spheres can touch a thirteenth and no more (Schütte and van der Waerden 1952), and twelve atoms round one make the icosahedral cluster of liquids and metals (Frank 1952). We set thirteen polyhedral thirteens side by side, each a centre and twelve around it. Six are golden, on the icosahedron's five-fold axes: the Icosa-13 (Lotus seed), DESHI, the Dodeca-13, UNITY with its closed form VENUS, the DICTO-Star and the Bodhi seed. Seven are periodic, a cell and twelve neighbours in a honeycomb: the rhombic dodecahedron (RD 13, with its rounded form the DICTO Spheroid-13 RD), DICTO's skewed RD, the Bain RD, the golden Bilinski dodecahedron, the regular-hexagon elongated dodecahedron, the truncated octahedron (4 + 5 + 4, the centre seen through two square windows) and the DICTO Jewel (with its stella octangulas). For each we give how the twelve touch the centre and one another, and the kits that fill between them: 24 kits from 13 to 213 pieces, every one checked piece by piece and buildable from the centre out. Two golden clusters close exactly into a $\varphi^2$-times larger copy of their own form. Every periodic block fills space by translation alone, and all seven are the same tiling in different shapes: under a linear change of coordinates each neighbourhood is the face-centred cubic one, whose radius-one perfect code is known (Kovačević and Tan 2018; explicitly $x_1 + 3x_2 + 9x_3 \equiv 0 \bmod 13$, Kovačević 2026), its weights Singer's difference set $\{0, 1, 3, 9\} \bmod 13$ (Singer 1938).

## 1. Twelve around one

Each cluster is a centre and twelve copies or neighbours around it. In the golden clusters the twelve lie on the twelve five-fold axes of the icosahedron; in the periodic ones they are twelve face neighbours of the centre in a space-filling honeycomb. Edge lengths are those of the pieces unless stated; $\varphi = (1 + \sqrt{5})/2$. In every cluster the centre is shown gold, and only the centre.

| Cluster | Centre | Twelve | Distance | Centre–twelve | Among the twelve |
|---|---|---|---|---|---|
| **Golden** | | | | | |
| Icosa-13 (Lotus seed) | icosahedron, edge 1/φ | icosahedra | 1.5388 | 12 at a corner | 30 along an edge |
| DESHI | icosahedron, edge 1/φ | icosahedra | 1.5388 | 12 at a corner | 30 along an edge |
| Dodeca-13 | dodecahedron | dodecahedra | 2.2270 | 12 by a whole face | 30 along an edge |
| UNITY / VENUS | icosidodecahedron | J11 | 1.8882 | 12 by a whole face | 30 at a corner |
| DICTO-Star | icosidodecahedron | dodecahedra | 2.4899 | 12 by a whole face | 30 along an edge |
| Bodhi seed | triacontahedron, edge 1/φ | triacontahedra | $\varphi^2 = 2.6180$ | 12 at a corner | 30 by a whole face |
| **Periodic** (all twelve by a whole face) | | | | | |
| RD 13 | rhombic dodecahedra | | √2 | 12 | 24 by a face, 12 at a corner |
| DICTO skewed RD 13 | DICTO skewed rhombic dodecahedra | | 1.1756–1.7321 | 12 | 24 by a face, 12 at a corner |
| Bain RD 13 | Bain rhombic dodecahedra | | 1.4142–1.7321 | 12 | 24 by a face, 12 at a corner |
| Bilinski 13 | Bilinski dodecahedra (golden rhombi) | | 1.0515–1.7013 | 12 | 24 by a face, 12 at a corner |
| Regular-hexagon ED 13 | regular-hexagon elongated dodecahedra | | 1.7321–2.4495 | 12 (4 across hexagons) | 24 by a face, 8 along an edge |
| Truncated octahedron 13 | truncated octahedra, 4 + 5 + 4 | | 2.4495–2.8284 | 12 (8 hexagons, 4 squares) | 28 by a face |
| DICTO Jewel 13 | DICTO Jewels | | 2.2882 | 12, window to window | 24 by a face |

DESHI is the Lotus seed's thirteen icosahedra on their own. In every golden cluster the thirty contacts among the twelve follow the icosahedron's thirty edges; in RD 13 and its stretched forms the twenty-four follow the cuboctahedron's edges. The regular-hexagon ED 13 and the truncated octahedron 13 both stand in three layers, 4 + 5 + 4; the truncated octahedron's top and bottom squares stay open, two windows onto the gold centre.

## 2. What fills between

| Cluster | Kit | Pieces | Kinds | Volume | Of its hull |
|---|---|---|---|---|---|
| Icosa-13 | pieces | 113 | 13 icosahedra, 60 AXE, 20 FUJI, 20 Cleo | 39.148929 | exactly 1 (icosahedron of edge $\varphi^2$) |
| | 63-piece build | 63 | dodecahedron, 12 J11, 30 AXE, 20 FUJI | 39.148929 | exactly 1 |
| | units | 13 | centre and 12 Hasu units | 39.148929 | exactly 1 |
| Dodeca-13 | pieces / units / stars | 63 / 13 / 33 | 13 dodecahedra with wedges and needles, or units, or 20 stars | 106.0585 / 106.0585 / 112.4126 | 0.742 / 0.742 / 0.786 |
| UNITY / VENUS | open / closed | 13 / 33 | icosidodecahedron, 12 J11, and 20 VAJRA | 36.3978 / 46.9352 | 0.643 / 0.829 |
| DICTO-Star | | 33 | icosidodecahedron, 12 dodecahedra, 20 J63 | 131.3367 | 0.724 |
| DESHI | | 13 | 13 icosahedra | 26.6954 | 0.682 |
| Bodhi seed | seed / Kit A / Kit B | 13 / 193 / 213 | 13 triacontahedra; Vajra (fine), Chatra, Mudra; Vajra (fine), Crown Jewels, Diadems L and R | 150.6347 / 220.9069 / 220.9070 | 0.703 / 1 / 1 (triacontahedron of edge $\varphi^2$) |
| RD 13 | plain; Spheroid-13 RD 0.15 and 0.35; free-centre variant | 13 each | rhombic dodecahedra; a true rhombic dodecahedron at the centre and two outer kinds | 26 each | 0.672; 0.886 and 0.793; 0.897 |
| Five other periodic | centre and twelve | 13 each | the shape itself | 13 cells | 0.672 (RD forms), 0.690 (ED), 0.719 (TO) |
| DICTO Jewel 13 | alone / with stellas | 13 / 19 | 13 DICTO Jewels, and 6 stella octangulas | 82.6033 / 95.3115 | 0.557 / 0.642 |

"Of its hull" is the kit's volume over that of its convex hull. The Icosa-13 and the Bodhi seed close exactly into a larger copy of their own form, $\varphi^2$ times the twelve (the Bodhi kits' last digits reflect coordinates stored to five places). The Dodeca-13, UNITY, VENUS and the DICTO-Star are filled solids with dimples by design; DESHI and the bare Bodhi seed are frames.

## 3. The seven periodic thirteens are one tiling

A cell of a lattice honeycomb and its twelve face neighbours fill space by translation exactly when the thirteen lattice points are a complete set of representatives of a sublattice of index 13. For every periodic member, 8 of the 183 such sublattices work, and seven blocks were checked free of overlap. The reason is shared: under a linear change of coordinates, the twelve neighbours of the rhombic dodecahedron, of DICTO's skewed RD, the Bain RD and the Bilinski dodecahedron, of the regular-hexagon elongated dodecahedron, and of the truncated octahedron's 4 + 5 + 4 are all exactly the twelve neighbours of the face-centred cubic lattice, and the DICTO Jewels sit on that lattice itself. So all seven are one tiling, the radius-one perfect code of the $A_3$ lattice (Kovačević and Tan 2018), written explicitly as $\{x : x_1 + 3x_2 + 9x_3 \equiv 0 \pmod{13}\}$ by Kovačević (2026); its weights 0, 1, 3, 9 are Singer's perfect difference set modulo 13 from the projective plane of order 3 (Singer 1938), every nonzero residue a difference of two of them exactly once. The DICTO Jewel 13 blocks cover the Jewel sites of the Stella–Jewel checkerboard in the same way, the stella octangulas filling the remaining cells.

Seven shapes wear the one tiling differently. The rhombic dodecahedron's own block, rounded with flat faces and kept keyed around a true rhombic dodecahedron, is the DICTO Spheroid-13 RD, described in its own paper (DICTO, DICTO Spheroid-13 RD, 2026, doi:10.5281/zenodo.23332821).

## 4. Kits to build

All 24 kits are buildable piece by piece in the Polyhedraverse app (DICTO's pieces). Each kit is a plan, every piece's shape, place and turn about the centre; choosing a kit puts its centre down, the plan is lined up with the build, and the next layer still open is shown as outlines, from the centre out. The kits reuse catalogue solids wherever a piece is exactly one (dodecahedra, icosahedra, J11, J63, the icosidodecahedron, the triacontahedra, the rhombic dodecahedra, the elongated dodecahedron, the truncated octahedron, the DICTO Jewel, the stella octangula) and add DICTO's pieces where not (among them AXE, FUJI, Cleo, Hasu, VAJRA, Vajra (fine), Chatra, Mudra, Crown Jewel, Diadem L and R, the Bilinski dodecahedron and the Spheroid-13 RD pieces). The DICTO ATOM-13×13 world of the Kaleidohedra app shows every kit with pull-apart, shell-by-shell and see-through views, and the periodic ones as seven blocks filling space.

## 5. Checks

Every piece of every kit: closed, consistently wound, faces flat (worst $1.5 \times 10^{-5}$ from five-place coordinates in the Bodhi kits; $10^{-13}$ or better elsewhere). Kit volumes as tabled; the closed kits equal their hulls. Every placement of every kit plan reproduces its piece exactly (corners within $10^{-4}$), and every kit builds to completion following its outlines from a centre placed at an arbitrary turn and place. For each periodic member: 13 points distinct modulo 8 of the 183 sublattices; seven blocks with no point in two cells (20,000 samples); the neighbourhood's linear equivalence to the face-centred cubic one found as an explicit integer matrix. Scripts: scripts/dump-thirteens.mjs, build-thirteens-data.py, build-kits13.py, kit-shapes.py; papers/thirteens/fig.

## 6. Relation to known work

Twelve spheres round one and the thirteen-sphere problem: Schütte and van der Waerden (1952). The thirteen-atom icosahedral cluster: Frank (1952); its shells as a dense non-crystallographic packing: Mackay (1962). Icosahedral scaling and packing: Gray (2002). Clusters of triacontahedra: Kabai. Parallelohedra, the solids that tile by translation: Fedorov (1885), Grünbaum (2010). The periodic thirteens' common tiling is the $A_3$ radius-one perfect code (Kovačević and Tan 2018; Kovačević 2026; Singer 1938), within the theory of perfect codes in lattices (Golomb and Welch 1970; Etzion 2011; Campello et al. 2015; Gravier, Mollard and Payan 1998). We found no description of these thirteen clusters side by side, of the golden ones' kits beyond DICTO's own papers, of the truncated octahedron's 4 + 5 + 4 or the DICTO Jewel 13 as blocks, or of the seven periodic shapes read as one tiling. The search is not exhaustive.

## Figures

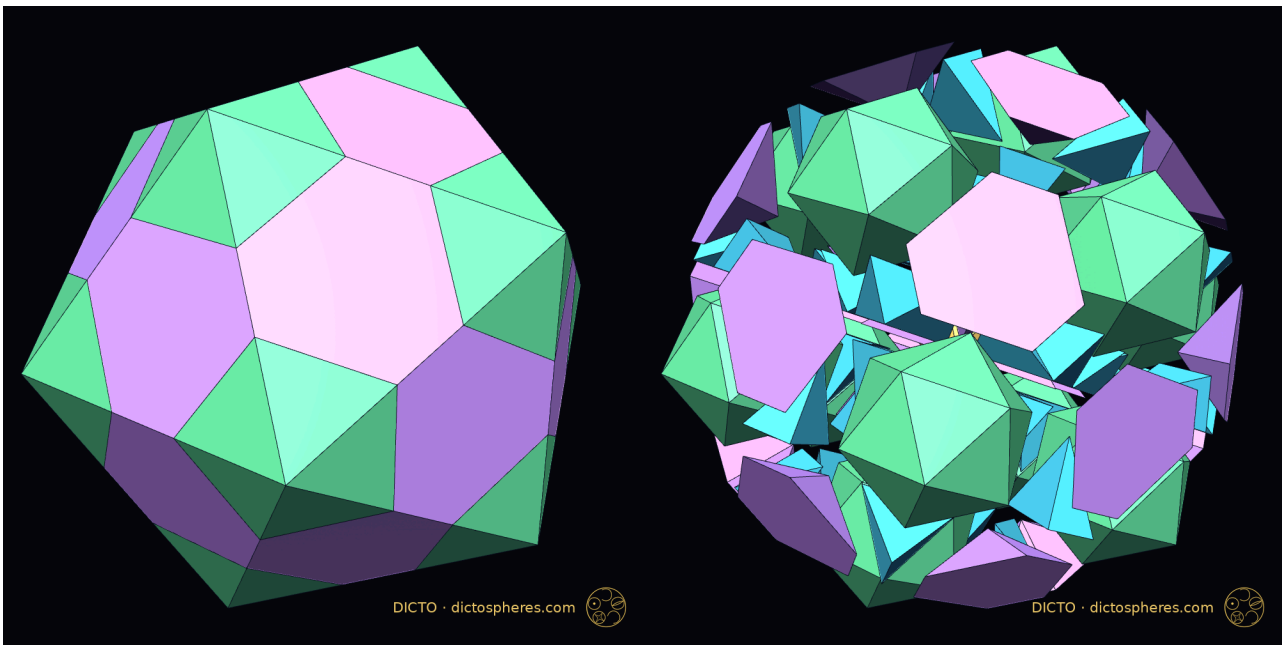



**Figure 1.** Icosa-13 (Lotus seed), 113 pieces, closed and pulled apart.

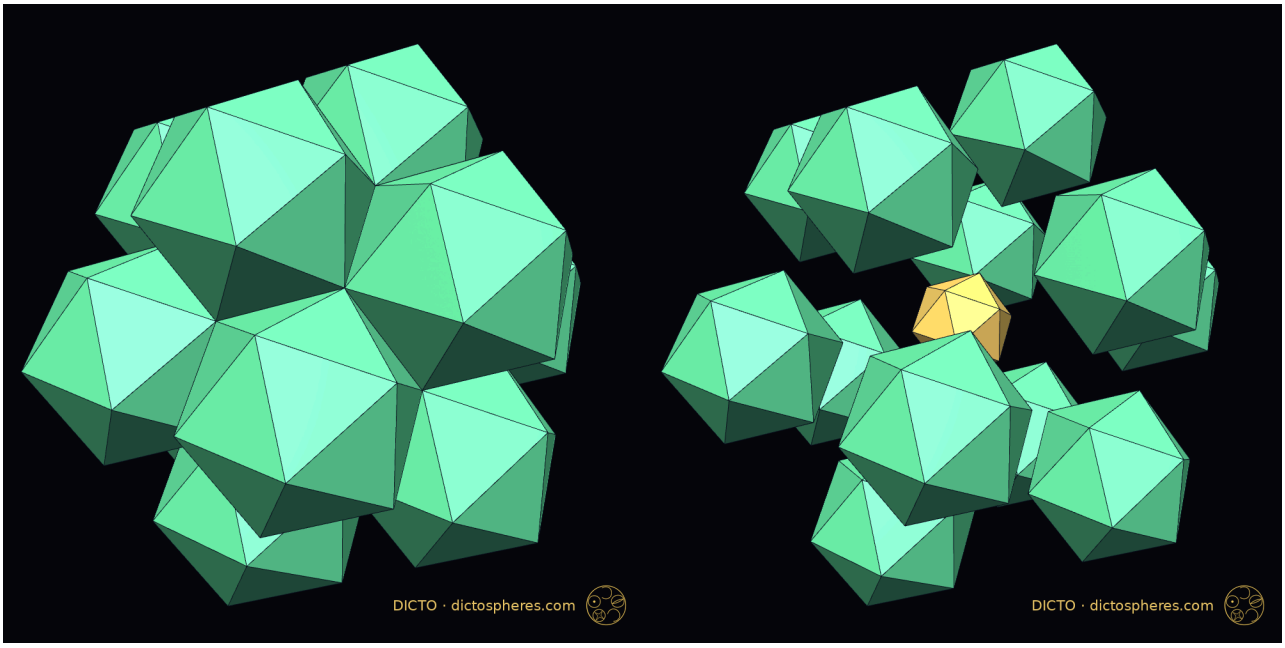



**Figure 2.** DESHI: the Lotus seed's thirteen icosahedra alone.

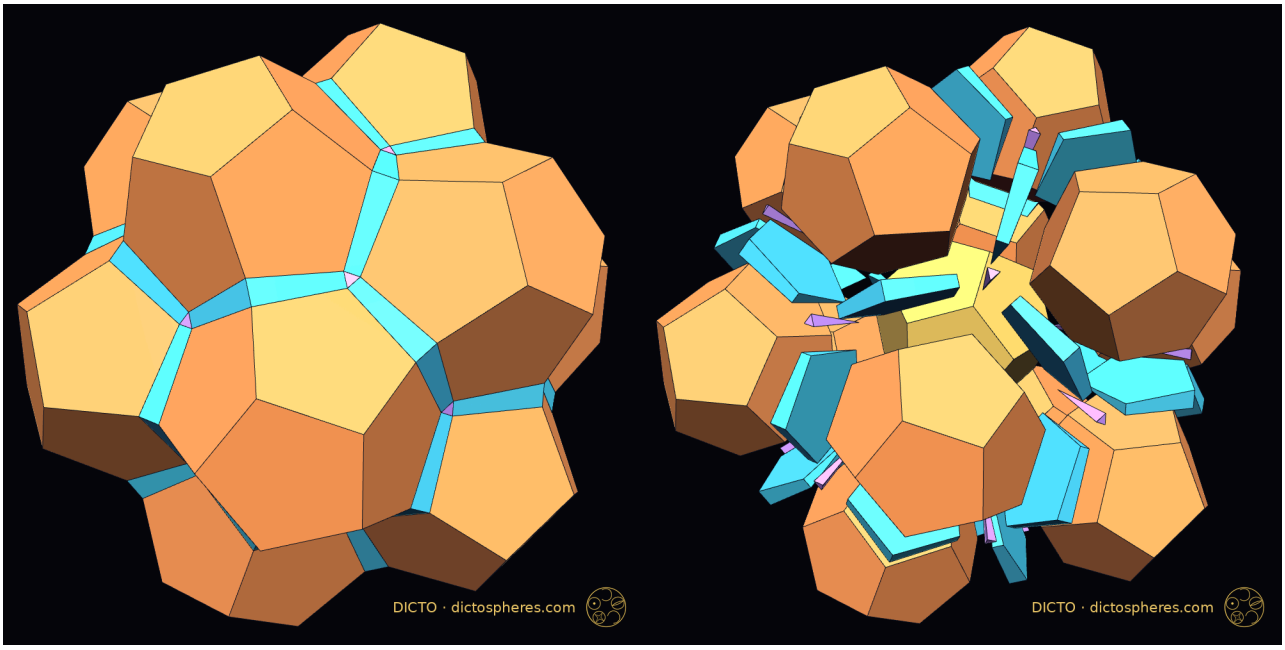



**Figure 3.** Dodeca-13: thirteen dodecahedra, 30 wedges, 20 needles.

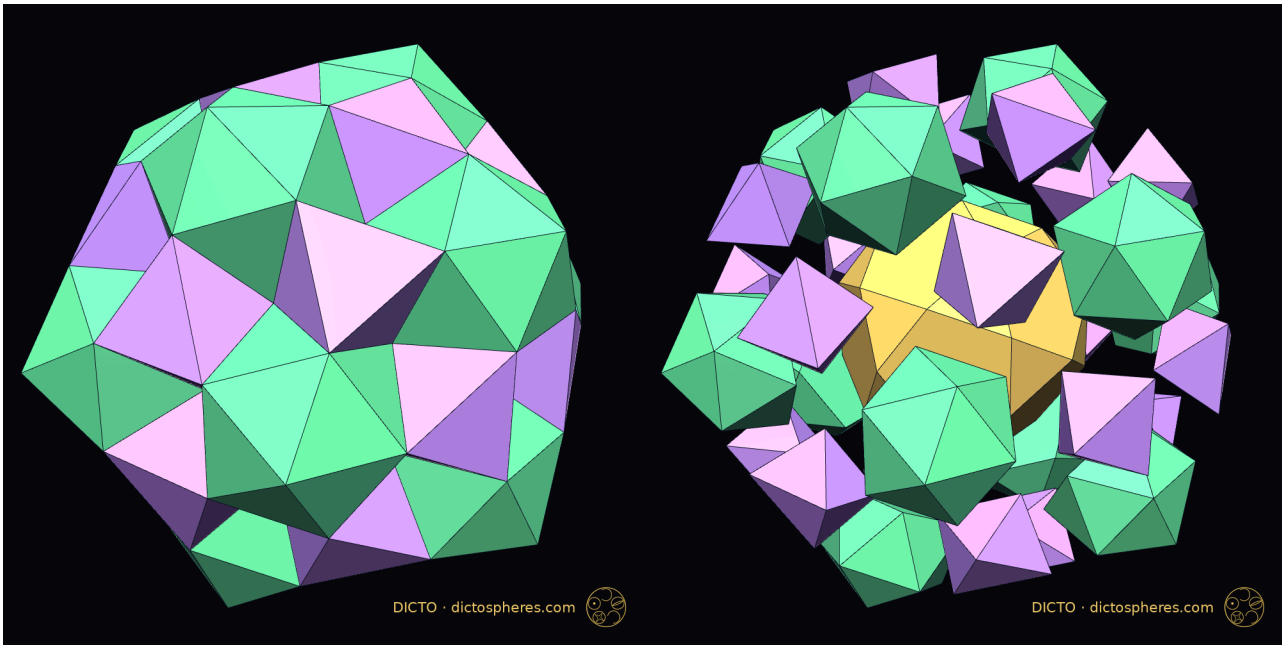



**Figure 4.** VENUS: UNITY (icosidodecahedron and 12 J11) closed by 20 VAJRA.

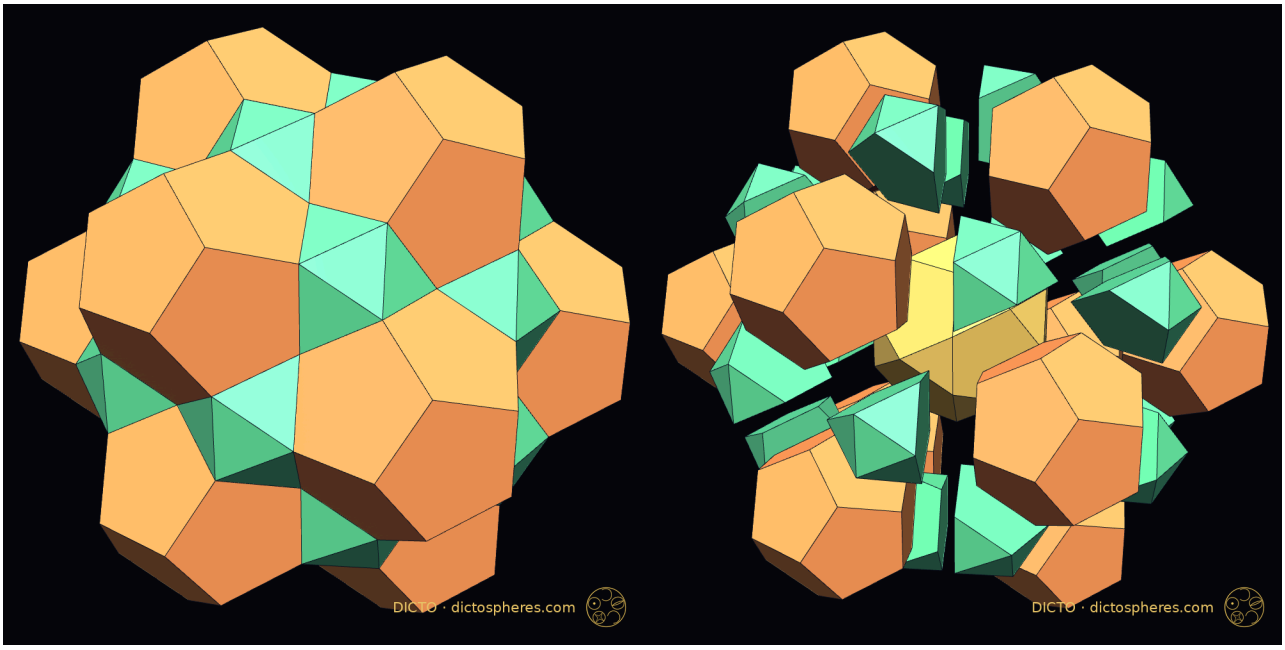



**Figure 5.** DICTO-Star: icosidodecahedron, 12 dodecahedra, 20 J63.

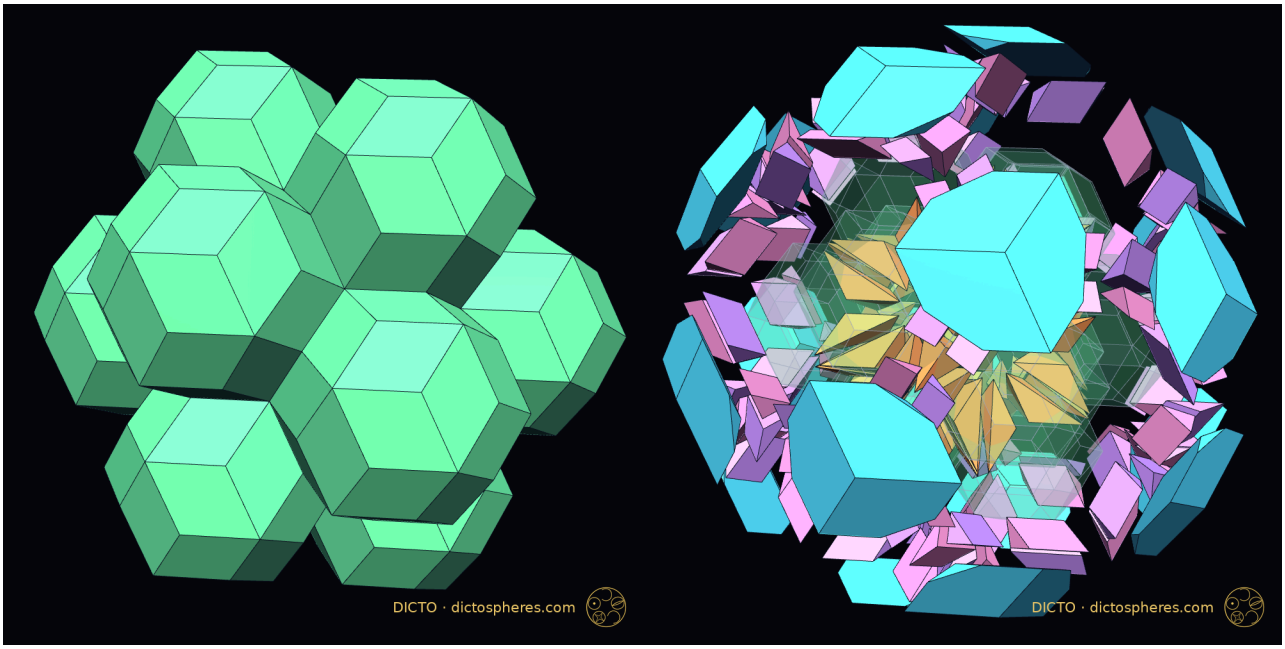



**Figure 6.** The Bodhi seed, and Kit B pulled apart with the twelve see-through.

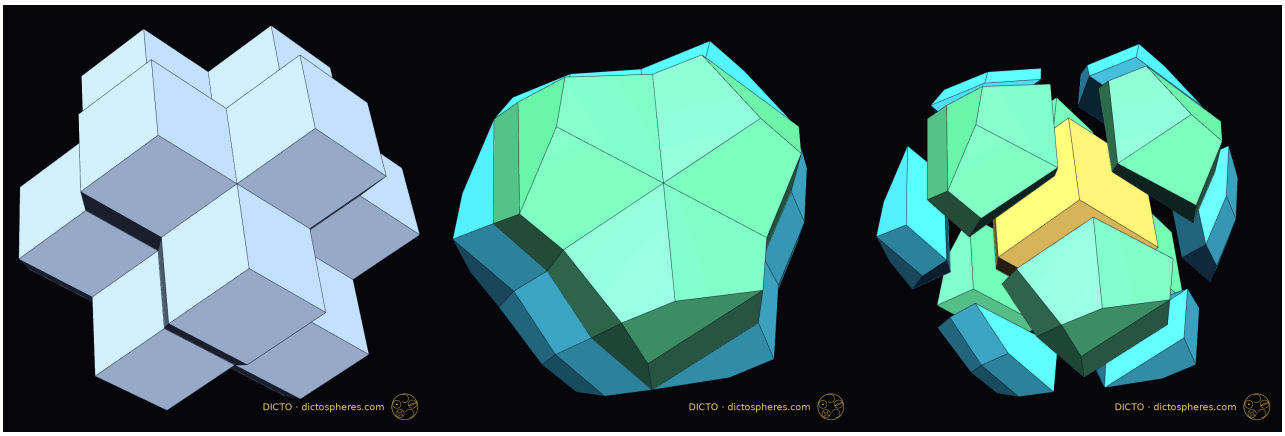



**Figure 7.** RD 13 and the DICTO Spheroid-13 RD, whole and pulled apart, a true rhombic dodecahedron at its heart.

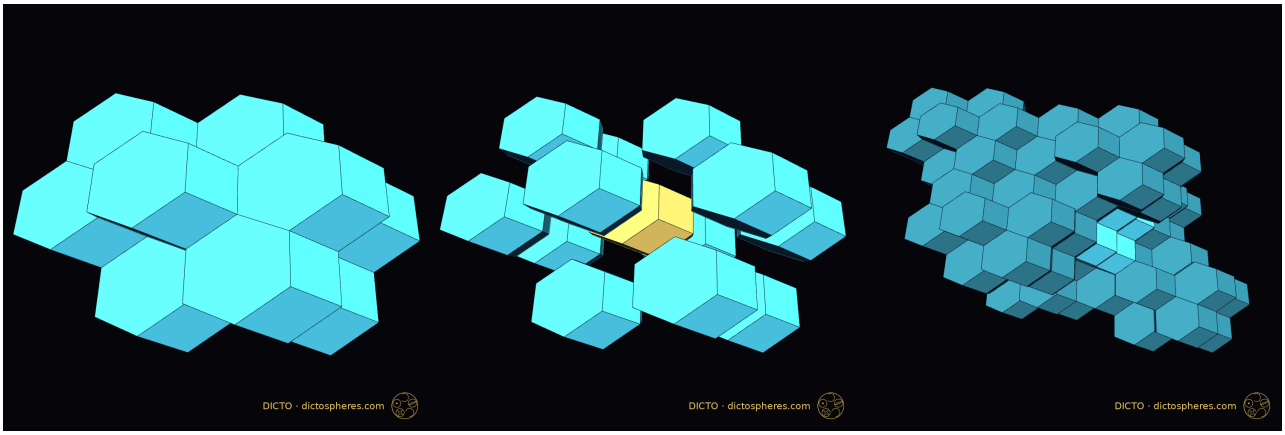



**Figure 8.** Regular-hexagon ED 13: whole, pulled apart, seven blocks.

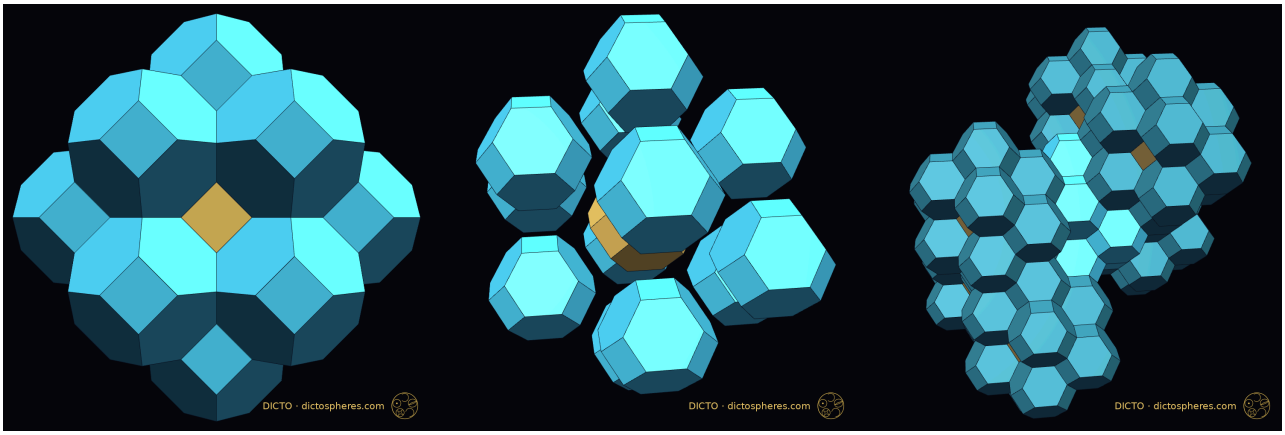



**Figure 9.** Truncated octahedron 13, 4 + 5 + 4: from above, straight down its window, the gold centre's square at the middle; pulled apart; seven blocks.

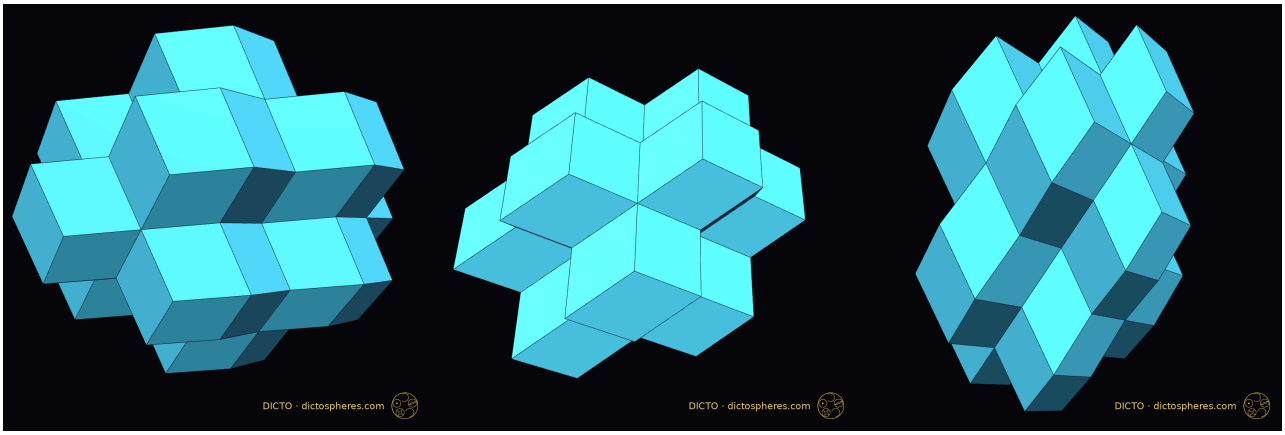



**Figure 10.** DICTO skewed RD 13, Bain RD 13 and Bilinski 13.

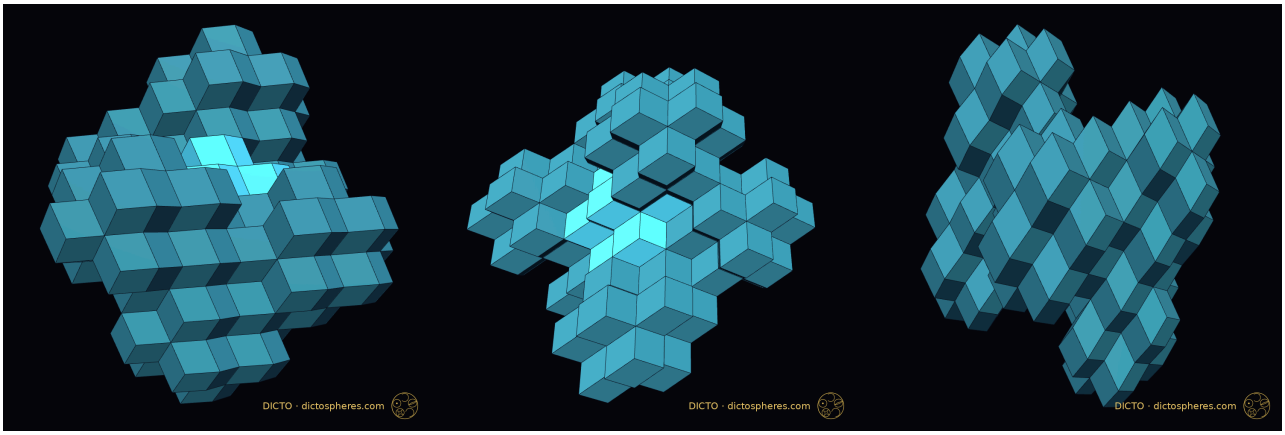



**Figure 11.** The same three as seven blocks filling space: one tiling, three shapes.

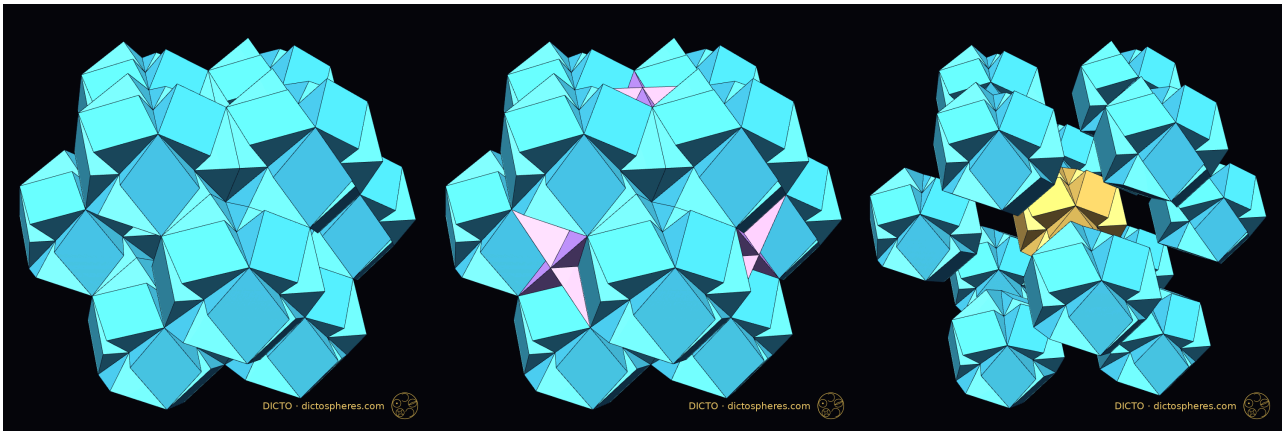



**Figure 12.** DICTO Jewel 13: the thirteen Jewels, with the 6 stella octangulas, pulled apart.

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